

HTX9000 CALIBRATION PROCEDURE

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Purpose: This document describes how to calibrate the HTX9000 automatic USB connected load box and how to read back the calibration data.

Items needed:

HTX9000 USB controlled load box

USB cable with “mini-B” connector.

Power supply able to produce at least 2.5 amps of current at least 5 VDC

Accurate current meter with Amps and milliamp ranges

Jumper cables.

Procedure:

- 1) Connect the load box to a PC via USB cable. It should power up, show its logo, and show the load setting. If the load box is uncalibrated, ignore the load setting on the display.
- 2) On the PC, determine the virtual COM port attached to the load box and open a terminal program on that COM port to communicate with the device.
- 3) In the terminal program, send the command “*IDN?” The load box should respond with its make and model information.
- 4) Set the power supply to 5VDC, connect it to the load box with the current meter between them.
- 5) Send the following commands to the load box, in the following order:
 - a. CALibrate:DATE (today’s date, ex: 4/26/13)
 - b. CALibrate:OUTPut:CURRent:DAC:RANGe:HI 65535
 - c. CALibrate:OUTPut:CURRent:VALue:RANGe:HI (meter reading, in amps. Ex: 2.4998A)
 - d. CALibrate:OUTPut:CURRent:DAC:RANGe:LO 65535
 - e. CALibrate:OUTPut:CURRent:VALue:RANGe:LO (meter reading, in mA. Ex: 24.99962mA)
- 6) While calibrating, the display will be locked to prevent changes on the load box’s control buttons.
- 7) After calibration, command the load box to various load currents with the following command: “SOURce:CURRent 1A” or “SOURce:CURRent 10mA” and observe the meter measuring the commanded current.
- 8) The Board can be given a version number or title limited to 16 characters such as “Steve’s LoadBox” with:
 - a. CALibrate:BOArdrev Steve’s LoadBox
- 9) Mark the meter as calibrated with a dated calibration label.